Creating assets

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Create assets and assign contacts to assets by using the GraphQL API.

Before you begin

Get a JSON Web Token (JWT) for your API call. For more information, see API authentication.

About this task

An asset represents a location, which is defined by geographic coordinates. Contacts are associated with each asset.
The URL includes a `{geo}` field, which is a 2-letter code that represents the region. For North America, the code is na.

Procedure

1. Create an asset by sending an `upsertAsset` event to the GraphQL API.

> **Note:**
> You use the same procedure to update an asset. To update an asset, provide the same payload and include the required updates.

URL	`https://api.ibm.com/infohub/run/graph/na`
Method	POST
Body-Type	GraphQL
Body	

```
mutation {
  upsertAsset(
    AssetEvent: {
      tenantId: "<tenantID>"
      timestampEventOccurred: "<2023-02-22T16:00:00Z>"
      eventDetails: {
        businessObject: {
          globalIdentifiers: [
            {
              name: "eis.geospatial.globalId",
              value: "<assetIdentifier>.STATIC"
            }
          ]
          status: "<assetStatus>"
          assetIdentifier: "<assetIdentifier>"
          assetType: "STATIC"
          currentLocationCoordinates: "<co-ordinates for example, 40.113, -99.295>"
          description: "<description>"
          contacts: [
            {
              globalIdentifiers:  [
                       {name: "eis.geospatial.globalId",
                        value: "{contact_ID}"}
                   ]
               }]
          location: {
            globalIdentifiers: [
                      {
                        name: "eis.geospatial.globalId",
                        value: "{location_ID}"
                      }
                   ]
          }
        }
      }
    })
}
```

- Include the following key value pairs in the header:

```
Authorization: Bearer {{your-JWT-token}}
x-ibm-client-id: infohub-{{your-tenant-id}}
```

- Add `tenantId` as a field.
- Assign an asset ID to `eventDetails.businessObject.globalIdentifiers`.
- Set the asset type to either `STATIC`.
- Set the `assetStatus` as active.
- Refer to the asset in the `globalIdentifiers` section. Match the value of the `assetIdentifier`field followed by a dot and then the value of the `assetType` field. For example, if the `assetIdentfier` is `asset123`, and the `assetType` is STATIC, then the `globalIdentifier` value is `asset123.STATIC`
- Define the geographic location of the asset in `currentLocationCoordinates`. For example, `currentLocationCoordinates: "40.113, -99.295"` is 40.113 latitude and -99.295 longitude.

> **Note:**
>
> If the location coordinates are not well-defined, the asset is not recognized by the Alerting application.

- Add one or more contacts to `eventDetails.contacts.globalIdentifiers`.
- To append more contacts to the asset, update the following code:

```
{
            globalIdentifiers:  [
                     {name: "eis.geospatial.globalId",
                      value: "{contact_ID"}
                ]
             }
```

Append the new contact ID, for example:

```
{
            globalIdentifiers:  [
                     {name: "eis.geospatial.globalId",
                      value: "{another_contact_ID}"}
                ]
             }
```

2. To find a list of assets, create a query.
   Use the GraphQL API to fetch the list of assets.

   The following examples show a sample query and the result that was returned with the asset data.

URL

```
https://api.ibm.com/infohub/run/graph/na
```

Query criteria

```
query getAsset {
  businessObjects(
    hint: { viewId: "graph" }
    advancedFilter: {
      AND: [
        {
          EQUALS: [
            { SELECT: "globalIdentifiers.name" }
            { VALUE: "eis.geospatial.globalId" }
          ]
        }
        {
          EQUALS: [
            { SELECT: "globalIdentifiers.value" }
            { VALUE: "{asset_ID.STATIC}" }
          ]
        }
      ]
    }
    simpleFilter: { type: Asset, tenantId: "{tenant_ID}" }
  ) {
    totalCount
    edges {
      object {
        ... on Asset {
          description
```

```
          assetIdentifier
          assetType
          currentLocationCoordinates
          contacts {
            edges {
              object {
                ... on Contact {
                  globalIdentifiers {
                    name
                    value
                  }
                  emailAddress
                  firstName
                  lastName
                }
              }
            }
          }
        }
      }
    }
  }
}
```

Query result

```
{{
        "data": {
                "businessObjects": {
                        "totalCount": 1,
                        "edges": [{
                                "object": {
                                        "description": "A description",
                                        "assetIdentifier": "asset123",
                                        "assetType": "STATIC",
                                        "currentLocationCoordinates": "26.32, -79.321",
                                        "contacts": {
                                                "edges": [{
                                                        "object": {
                                                                "globalIdentifiers": [{
                                                                        "name": "eis.contact.identifier",
                                                                        "value": "contact_ID"
                                                                }],
                                                                "emailAddress": "JD_email@email.com",
                                                                "firstName": "John",
                                                                "lastName": "Doe"
                                                        }
                                                }]
                                        }
                                }
                        }]
                }
        }
}
```

3. To delete an asset, use the `mutation` keyword and the `archiveBusinessEvent` method.
   Use the GraphQL API to delete the asset.

URL

```
https://api.ibm.com/infohub/run/graph/na
```

Body

```
mutation {
  archiveBusinessEvent(
    archiveEvent: {
      tenantId: "{tenant_ID)"
      eventDetails: {
        businessObject: {
          type: Asset
          globalIdentifiers: [
            { name: "eis.geospatial.globalId", value: "{asset_ID.STATIC}" }
          ]
        }
      }
    }
  )
}
```

- Include the following key value pairs in the header:

    - `'Authorization':'JWT token'`
    - `'x-ibm-client-id':'infohub-<tenantId>'`
  - Add `tenantId` as a field.
  - Set `eventDetails.businessObject.type` to `Asset`.
  - Specify the assets to delete in `eventDetails.businessObject.globalIdentifiers` .
4. To remove a contact from an asset, use the `upsertAsset` keyword.
   Use the GraphQL API to remove the contact from the asset.

URL	`https://api.ibm.com/infohub/run/graph/na`
Body	

```
mutation {
   upsertAsset (AssetEvent: {
       tenantId: "{tenant_ID}",
       timestampEventOccurred: "2023-02-24T16:00:00Z",
       eventDetails: {
           businessObject: {
               globalIdentifiers: [
                   {name: "eis.geospatial.globalId",
                    value: "{asset_ID}.{assetType}"}
               ]
               assetIdentifier: "{asset_ID},
               assetType: "STATIC",
               currentLocationCoordinates: "{for example, 26.32, -79.321}",
               description: "{Description}"
           },
           remove: {
               contacts: {
                   globalIdentifiers: [
                       {name: "eis.geospatial.globalId",
                        value: "{another_contact_ID}"}
                   ]
               }
           }
       }
   })}
```

  - Include the following key value pairs in the header:
    - `'Authorization':'JWT token'`
    - `'x-ibm-client-id':'infohub-<tenantId>'`
  - Add `tenantId` as a field.
  - Put a `remove` event at the same level as `businessObject`. The contact to remove is defined under `contacts`.
  - Specify the assets to delete in `eventDetails.businessObject.globalIdentifiers` .

Parent topic:

→ Setting up alerts by using APIs